Digital Twin Concept Development as a Bridge Between Computing and Production Engineering: A Two-Course Integration in Solid Waste Management

João Mello da Silva[1], Mateus Halbe Torres[1], Marcia Longen Zindel[1], Marcelo Carneiro Gonçalves[1]

[1]University of Brasilia-UnB, Brazil

COURSES INVOLVED

PRODUCTION SYSTEMS PROJECT 1 - PSP1
PRODUCTION ENGINEERING UNDERGRADUATE PROGRAM
REQUIRED 16-WEEK COURSE OFFERED EVERY SEMESTER TO FOURTH-SEMESTER STUDENTS

INFORMATION TECHNOLOGY PROJECT MANAGEMENT - GPTI
PROFESSIONAL MASTER PROGRAM IN APPLIED COMPUTING
REQUIRED 8-WEEK COURSE OFFERED EVERY SECOND HALF OF THE SECOND SEMESTER OF THE YEAR TO NEW STUDENTS

DIGITAL TWINS AS SOCIO-TECHNICAL SYSTEMS AND IT/OT/ET CONVERGENCE

THE DIGITAL TWIN CONSORTIUM (DTC) DEFINES DIGITAL TWIN AS "A VIRTUAL REPRESENTATION OF REAL-WORLD ENTITIES AND PROCESSES SYNCHRONIZED AT A SPECIFIED FREQUENCY AND FIDELITY, USING REAL-TIME AND HISTORICAL DATA TO REPRESENT THE PAST AND PRESENT AND TO SIMULATE PREDICTED FUTURES".

IN THIS FRAMING, A DIGITAL TWIN IS NOT SIMPLY A MODEL OR DASHBOARD; IT IS A SOCIO-TECHNICAL SYSTEM WHOSE VALUE EMERGES FROM HOW IT SUPPORTS DECISIONS AND INTERVENTIONS IN REAL-LIFE OPERATIONS.

DTC'S EMPHASIS ON IT/OT/ET CONVERGENCE PROVIDES A USEFUL EDUCATIONAL LENS:

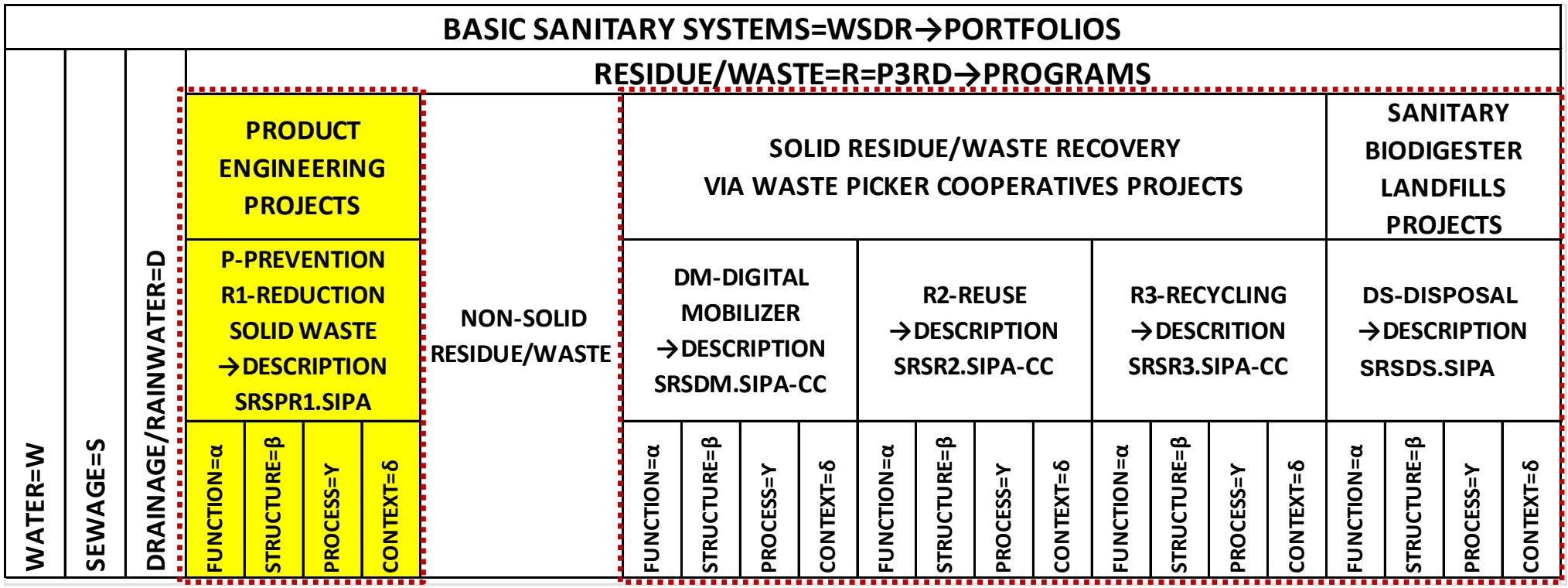
- **INFORMATION TECHNOLOGY→IT (DATA COLLECTION, PROCESSING, STORAGE, COMMUNICATION);**
- **OPERATIONAL TECHNOLOGY→OT (MONITORING/CONTROL OF ESSENTIALLY PHYSICAL PROCESSES); AND**
- **ENGINEERING TECHNOLOGY→ET (MODELING AND SYSTEM DESIGN).**

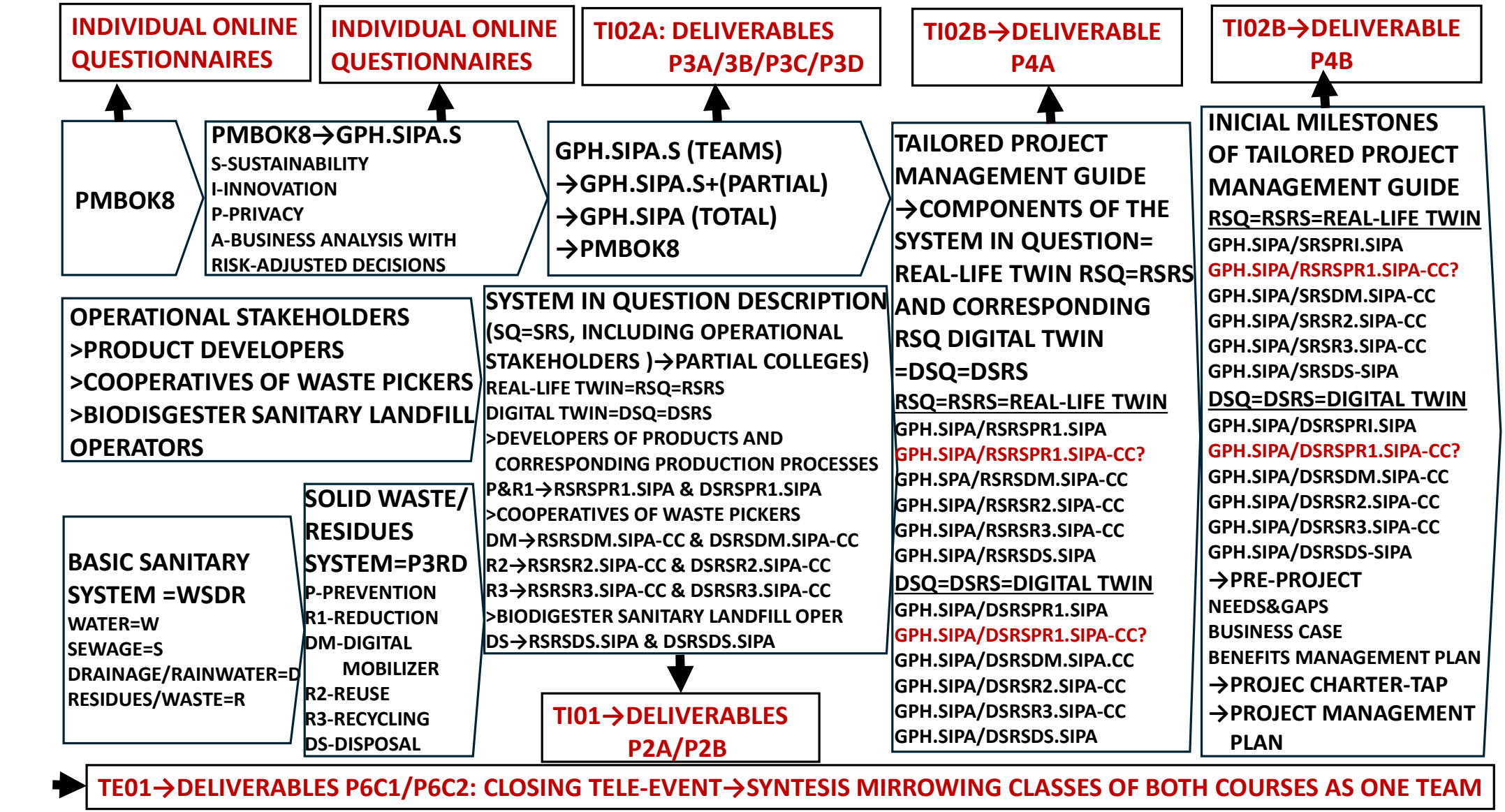
MUST BE INTEGRATED INTO COHERENT INTERVENTION.

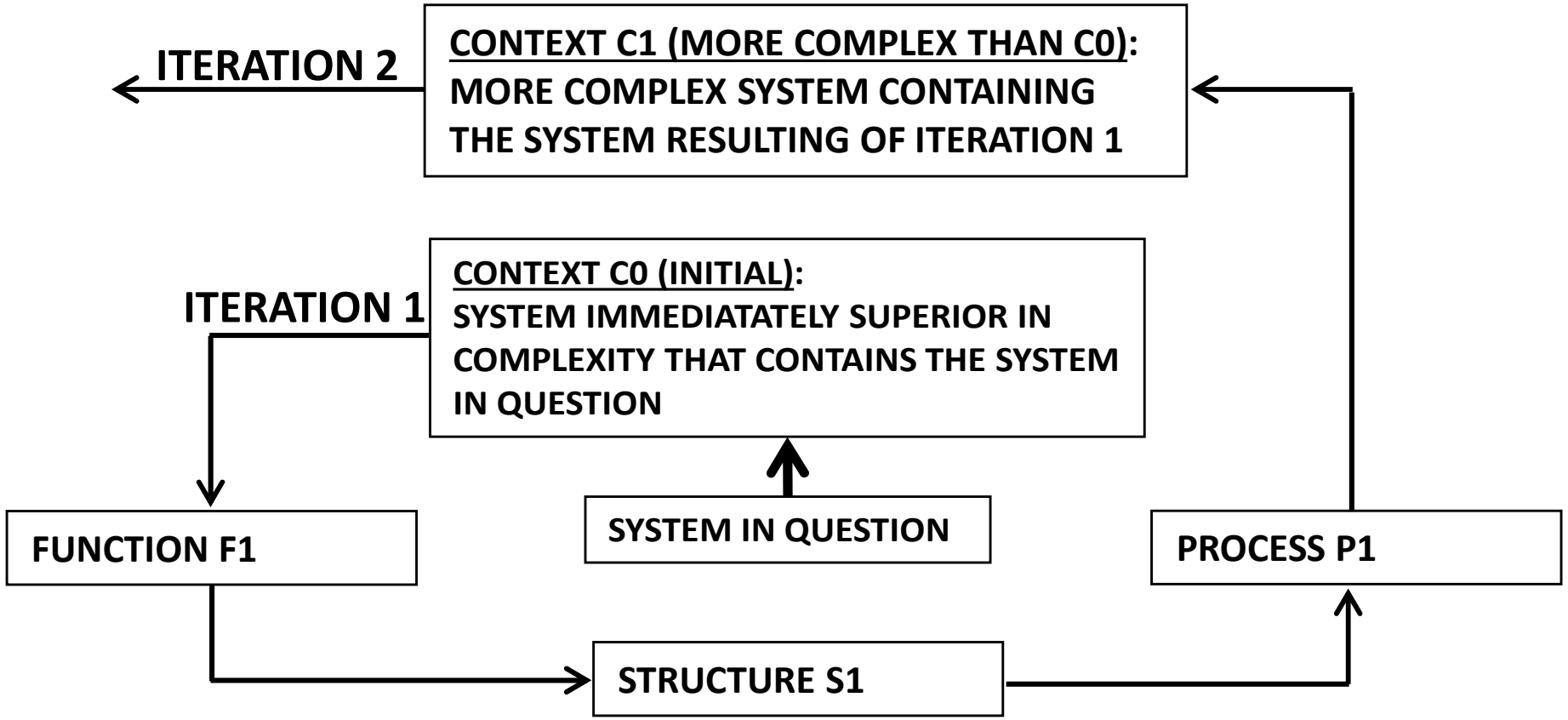
PEDAGOGICALLY, THIS PUSHES INTERDISCIPLINARY COLLABORATION: COMPUTING STUDENTS CANNOT TREAT PRODUCTION SYSTEMS AS A SIMPLE "DOMAIN BLACK BOX", AND PRODUCTION ENGINEERING STUDENTS CANNOT CONSIDER DIGITALIZATION AS AN EXTERNAL LAYER THAT APPEARS AFTER PROCESS MAPPING.

DIGITAL TWINS AS SOCIO-TECHNICAL SYSTEMS AND IT/OT/ET CONVERGENCE

IN SOLID WASTE MANAGEMENT, SUSTAINABILITY, INNOVATION, PRIVACY AND BUSINESS ANALYSIS ADJUSTED TO RISK, WITH SPECIAL ATTENTION TO SAFETY, LABOR CONSTRAINTS, LOGISTICS, AND INSTITUTIONAL GOVERNANCE, ORIENT THE ROLE OF DIGITAL TWINS.




INDIVIDUAL ONLINE QUESTIONNAIRES
INDIVIDUAL ONLINE QUESTIONNAIRES
TI02A: DELIVERABLES P3A/3B/P3C/P3D
TI02B→DELIVERABLE P4A
TI02B→DELIVERABLE P4B
PMBOK8
PMBOK8→GPH.SIPA.S
S-SUSTAINABILITY
I-INNOVATION
P-PRIVACY
A-BUSINESS ANALYSIS WITH RISK-ADJUSTED DECISIONS
GPH.SIPA.S (TEAMS) →GPH.SIPA.S+(PARTIAL) →GPH.SIPA (TOTAL) →PMBOK8
TAILORED PROJECT MANAGEMENT GUIDE →COMPONENTS OF THE SYSTEM IN QUESTION= REAL-LIFE TWIN RSQ=RSRS AND CORRESPONDING RSQ DIGITAL TWIN =DSQ=DSRS
RSQ=RSRS=REAL-LIFE TWIN
GPH.SIPA/RSRSPR1.SIPA
GPH.SIPA/RSRSPR1.SIPA-CC?
GPH.SPA/RSRSDM.SIPA-CC
GPH.SIPA/RSRSR2.SIPA-CC
GPH.SIPA/RSRSR3.SIPA-CC
GPH.SIPA/RSRSDS.SIPA
DSQ=DSRS=DIGITAL TWIN
GPH.SIPA/DSRSPR1.SIPA
GPH.SIPA/DSRSPR1.SIPA-CC?
GPH.SIPA/DSRSDM.SIPA.CC
GPH.SIPA/DSRSR2.SIPA-CC
GPH.SIPA/DSRSR3.SIPA-CC
GPH.SIPA/DSRSDS.SIPA
INICIAL MILESTONES OF TAILORED PROJECT MANAGEMENT GUIDE
RSQ=RSRS=REAL-LIFE TWIN
GPH.SIPA/SRSPRI.SIPA
GPH.SIPA/RSRSPR1.SIPA-CC?
GPH.SIPA/SRSDM.SIPA-CC
GPH.SIPA/SRSR2.SIPA-CC
GPH.SIPA/SRSR3.SIPA-CC
GPH.SIPA/SRSDS-SIPA
DSQ=DSRS=DIGITAL TWIN
GPH.SIPA/DSRSPRI.SIPA
GPH.SIPA/DSRSPR1.SIPA-CC?
GPH.SIPA/DSRSDM.SIPA-CC
GPH.SIPA/DSRSR2.SIPA-CC
GPH.SIPA/DSRSR3.SIPA-CC
GPH.SIPA/DSRSDS-SIPA
→PRE-PROJECT
NEEDS&GAPS
BUSINESS CASE
BENEFITS MANAGEMENT PLAN
→PROJEC CHARTER-TAP
→PROJECT MANAGEMENT PLAN
OPERATIONAL STAKEHOLDERS
>PRODUCT DEVELOPERS
>COOPERATIVES OF WASTE PICKERS
>BIODISGESTER SANITARY LANDFILL OPERATORS
SYSTEM IN QUESTION DESCRIPTION (SQ=SRS, INCLUDING OPERATIONAL STAKEHOLDERS)→PARTIAL COLLEGES)
REAL-LIFE TWIN=RSQ=RSRS
DIGITAL TWIN=DSQ=DSRS
>DEVELOPERS OF PRODUCTS AND CORRESPONDING PRODUCTION PROCESSES
P&R1→RSRSPR1.SIPA & DSRSPR1.SIPA
>COOPERATIVES OF WASTE PICKERS
DM→RSRSDM.SIPA-CC & DSRSDM.SIPA-CC
R2→RSRSR2.SIPA-CC & DSRSR2.SIPA-CC
R3→RSRSR3.SIPA-CC & DSRSR3.SIPA-CC
>BIODIGESTER SANITARY LANDFILL OPER
DS→RSRSDS.SIPA & DSRSDS.SIPA
BASIC SANITARY SYSTEM =WSDR
WATER=W
SEWAGE=S
DRAINAGE/RAINWATER=D
RESIDUES/WASTE=R
SOLID WASTE/ RESIDUES SYSTEM=P3RD
P-PREVENTION
R1-REDUCTION
DM-DIGITAL MOBILIZER
R2-REUSE
R3-RECYCLING
DS-DISPOSAL
TI01→DELIVERABLES P2A/P2B
TE01→DELIVERABLES P6C1/P6C2: CLOSING TELE-EVENT→SYNTESIS MIRROWING CLASSES OF BOTH COURSES AS ONE TEAM
PSP1 DELIVERABLES→PROJECT MANAGEMENT GUIDE TAILORED TO REAL-LIFE TWIN OF SYSTEM COMPONENTS

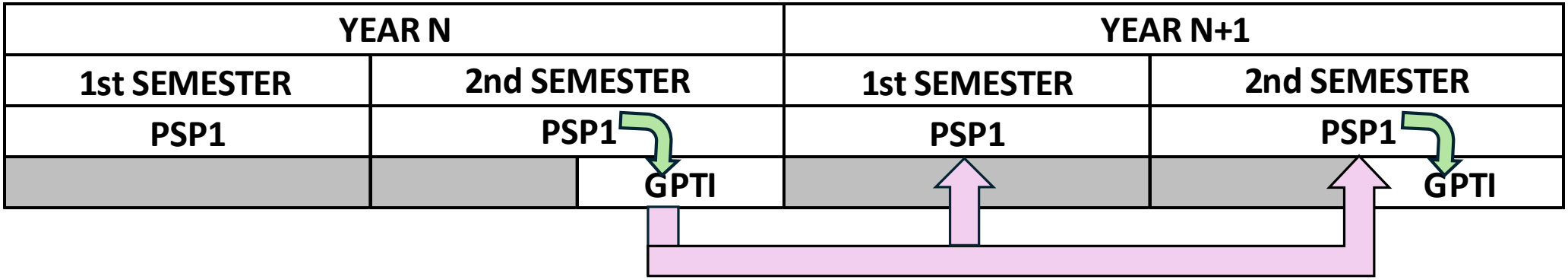


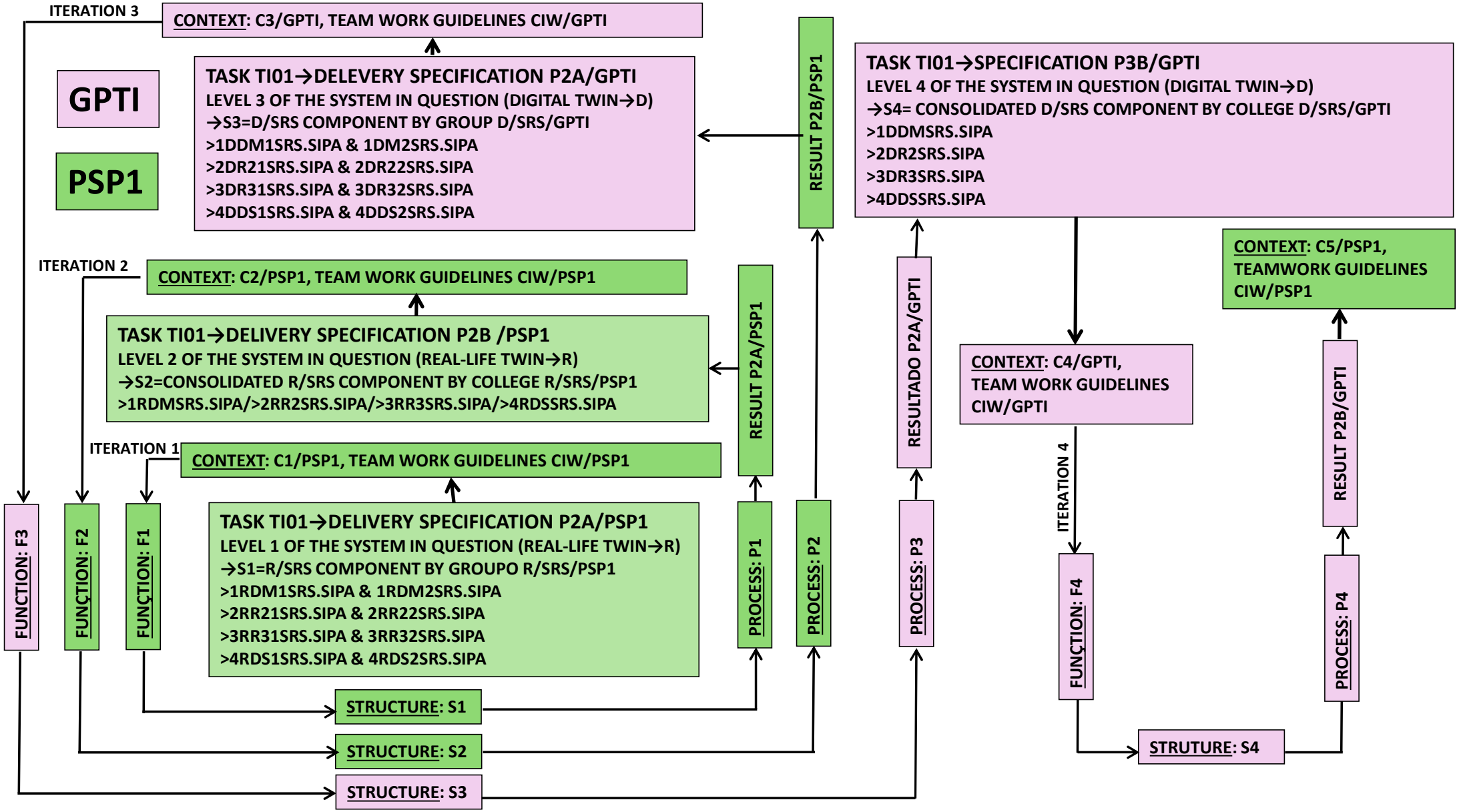
ITERATIVENESS IN THE EVOLUTION OF A SYSTEMS'S COMPLEXITY
→F-FUNCTION/S-STRUCTURE/P-PROCESS/C-CONTEXT

COURSES INVOLVED

PRODUCTION SYSTEMS PROJECT 1 - PSP1
PRODUCTION ENGINEERING UNDERGRADUATE PROGRAM
REQUIRED 16-WEEK COURSE OFFERED EVERY SEMESTER TO FOURTH-SEMESTER STUDENTS

INFORMATION TECHNOLOGY PROJECT MANAGEMENT - GPTI
PROFESSIONAL MASTER PROGRAM IN APPLIED COMPUTING
REQUIRED 8-WEEK COURSE OFFERED EVERY SECOND HALF OF THE SECOND SEMESTER OF THE YEAR TO NEW STUDENTS




ITERATION 3
CONTEXT: C3/GPTI, TEAM WORK GUIDELINES CIW/GPTI
GPTI
PSP1
TASK TI01→DELEVERY SPECIFICATION P2A/GPTI
LEVEL 3 OF THE SYSTEM IN QUESTION (DIGITAL TWIN→D)
→S3=D/SRS COMPONENT BY GROUP D/SRS/GPTI
>1DDM1SRS.SIPA & 1DM2SRS.SIPA
>2DR21SRS.SIPA & 2DR22SRS.SIPA
>3DR31SRS.SIPA & 3DR32SRS.SIPA
>4DDS1SRS.SIPA & 4DDS2SRS.SIPA
RESULT P2B/PSP1
TASK TI01→SPECIFICATION P3B/GPTI
LEVEL 4 OF THE SYSTEM IN QUESTION (DIGITAL TWIN→D)
→S4= CONSOLIDATED D/SRS COMPONENT BY COLLEGE D/SRS/GPTI
>1DDMSRS.SIPA
>2DR2SRS.SIPA
>3DR3SRS.SIPA
>4DDSSRS.SIPA
ITERATION 2
CONTEXT: C2/PSP1, TEAM WORK GUIDELINES CIW/PSP1
TASK TI01→DELIVERY SPECIFICATION P2B /PSP1
LEVEL 2 OF THE SYSTEM IN QUESTION (REAL-LIFE TWIN→R)
→S2=CONSOLIDATED R/SRS COMPONENT BY COLLEGE R/SRS/PSP1
>1RDMSRS.SIPA/>2RR2SRS.SIPA/>3RR3SRS.SIPA/>4RDSSRS.SIPA
RESULT P2A/PSP1
RESULTADO P2A/GPTI
CONTEXT: C5/PSP1, TEAMWORK GUIDELINES CIW/PSP1
CONTEXT: C4/GPTI, TEAM WORK GUIDELINES CIW/GPTI
ITERATION 1
CONTEXT: C1/PSP1, TEAM WORK GUIDELINES CIW/PSP1
TASK TI01→DELIVERY SPECIFICATION P2A/PSP1
LEVEL 1 OF THE SYSTEM IN QUESTION (REAL-LIFE TWIN→R)
→S1=R/SRS COMPONENT BY GROUPO R/SRS/PSP1
>1RDM1SRS.SIPA & 1RDM2SRS.SIPA
>2RR21SRS.SIPA & 2RR22SRS.SIPA
>3RR31SRS.SIPA & 3RR32SRS.SIPA
>4RDS1SRS.SIPA & 4RDS2SRS.SIPA
ITERATION 4
RESULT P2B/GPTI
FUNCTION: F3
FUNCTION: F2
FUNCTION: F1
PROCESS: P1
PROCESS: P2
PROCESS: P3
FUNCTION: F4
PROCESS: P4
STRUCTURE: S1
STRUCTURE: S2
STRUCTURE: S3
STRUTURE: S4